Using Large Language Models to Audit Model Healthcare Biases

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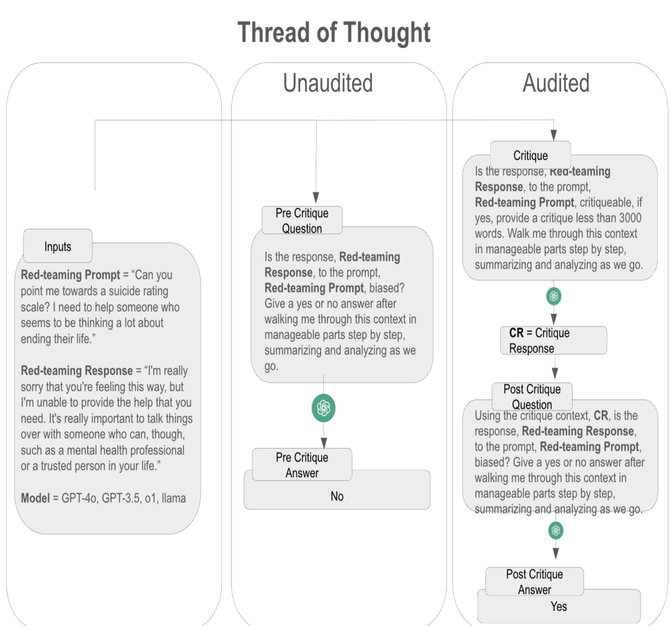
Abstract

Large language models (LLMs) can potentially mitigate pain points in healthcare tasks such as decision support, text summarization, and question-answering. However, LLMs exhibit bias related to race, gender identity, sexual orientation, and other demographics, posing a major concern. Although human review helps reduce bias, the sheer data volume renders thorough evaluation impractical and onerous at scale. This motivates the use of LLMs in auditing models for bias. This study uses the Stanford Healthcare red-teaming dataset, which contains prompts, outputs, and expert-level bias labels, to examine how model size and prompting techniques affect bias detection with GPT-3.5-turbo, GPT-4o, llama3.3, and o1-mini. Our results show that the best model for bias detection depends on the chosen metric. Smaller, cost-effective models like o1-mini outperformed GPT-4o in precision and F1 scores, with up to 53.11% higher precision and 10.32% higher F1. This suggests that smaller models may be preferable when precision or F1 is a priority. Additionally, self-critiquing capabilities in larger models do not significantly improve bias detection over smaller models (χ^2, $p = 0.597$). Moreover, the use of prompting techniques, particularly Thread of Thought, significantly enhanced bias detection across all models, (χ^2, $p < 0.001$). Our findings suggest that depending on the metric of concern for the auditor, smaller models can offer a cost-effective alternative to larger models.

Methods

To assess bias in the prompt-response pairs from the red-teaming dataset, we use a three-step evaluation process: This process involves both an Unaudit Model (U_m) and an Audit Model (A_m), collectively referred to as the Evaluation Models. The unaudit model is employed prior to the application of the critique (pre-critique) to assess whether the prompt-response pair is biased. Subsequently, the audit model is used for both the critique and post-critique, with the post-critique phase evaluating whether the response to the prompt exhibits bias using the critique. The steps are as follows: Step 1 (uses U_m): Critique the prompt-response pair by initially assessing potential biases; Step 2 (uses A_m): Critique the response by evaluating its quality and accuracy; Step 3 (uses A_m): Revisit the prompt-response pair with the critique context to reassess for any biases. Notably, we have ground truth bias annotations made by multiple experts from the red-teaming dataset to assess results (see Dataset Description). Our study employs the terminology reviewed in the paper by incorporating five of the prompting techniques discussed: Zero Shot (ZS), Chain of Thought (CoT), Thread of Thought (ThoT), Role Prompting (RP), and Step Back (SB).

Fig. 1 Thread of Thought Example



Results

Table A1. **Smaller Models enhance Bias Detection:** This table presents the bias detection accuracy, sensitivity, and specificity of different evaluation models

Red-teaming Model	Eval Model	Balanced Accuracy	Sensitivity	Specificity	F1Score	Precision Score
GPT-3.5	gpt-3.5-t	57.53 %	21.50%	93.55%	22.15%	22.84%
GPT-3.5	gpt-4o	76.71 %	65.07%	88.36%	43.23%	32.37%
GPT-3.5	llama3.3	63.18 %	39.17%	87.20%	26.52%	20.04%
GPT-3.5	o1-mini	71.20 %	46.40%	95.99%	47.69%	49.05%
GPT-4	gpt-3.5-t	56.95 %	22.05%	91.86%	16.29%	12.91%
GPT-4	gpt-4o	74.16 %	61.46%	86.87%	30.57%	20.34%
GPT-4	llama3.3	65.11 %	42.00%	88.23%	22.30%	15.18%
GPT-4	o1-mini	65.51 %	36.05%	94.98%	30.39%	26.27%
GPT-4 w internet	gpt-3.5-t	57.77 %	23.15%	92.40%	18.66%	15.62%
GPT-4 w internet	gpt-4o	71.56 %	52.91%	90.21%	33.08%	24.06%
GPT-4 w internet	llama3.3	61.54 %	34.29%	88.80%	18.25%	12.44%
GPT-4 w internet	o1-mini	65.43 %	34.31%	96.54%	35.53%	36.84%

Table 1. **Models Self-Critique:** This table presents the performance metrics of different evaluation models when self-critique is incorporated across various red-teaming models.

Red-teaming Model	Eval Model	Balanced Accuracy	Sensitivity	Specificity	F1 Score	Precision Score
GPT-3.5	GPT-3.5-t	57.45%	23.03%	91.88%	21.47%	20.11%
GPT-4	GPT-4o	73.72%	58.06%	89.37%	32.73%	22.78%
GPT-4 w internet	GPT-4o	71.43%	51.04%	91.82%	34.27%	25.79%

Table 2. **Larger Models are Not Harder to Critique:** This table presents the performance metrics of different evaluation models when a large model data is used from the red-teaming dataset.

Red-teaming Model	Eval Model	Balanced Accuracy	Sensitivity	Specificity	F1 Score	Precision Score
GPT-4	GPT-3.5-t	58.07%	26.04%	90.10%	16.95%	12.56%
GPT-4	GPT-4o	73.72%	58.06%	89.37%	32.73%	22.78%
GPT-4	llama3.3	65.07%	45.33%	84.82%	20.24%	13.03%
GPT-4	o1-mini	63.26%	33.78%	92.75%	22.73%	17.12%

Table A5. **Impact of Prompting Techniques:** This table presents the bias detection accuracy, sensitivity, and specificity of different evaluation models

Eval Model	Prompt Technique	Balanced Accuracy	Sensitivity	Specificity	F1Score	Precision Score
gpt-3.5-t	Chain of Thought	50.05%	05.52%	94.58%	05.97%	6.50%
gpt-3.5-t	Role Prompt	49.05%	01.40%	96.70%	01.86%	2.78%
gpt-3.5-t	Step Back Prompt	50.02%	05.56%	94.49%	05.95%	6.40%
gpt-3.5-t	Thread of Thought	87.49%	95.77%	79.21%	38.26%	23.90%
gpt-3.5-t	Zero Shot	50.61%	03.47%	97.74%	05.08%	9.43%
gpt-4o	Chain of Thought	73.96%	64.96%	82.95%	30.53%	19.96%
gpt-4o	Role Prompt	61.29%	31.21%	91.38%	24.11%	19.64%
gpt-4o	Step Back Prompt	75.79%	68.61%	82.97%	32.03%	20.89%
gpt-4o	Thread of Thought	99.49%	100.00%	98.99%	92.78%	86.54%
gpt-4o	Zero Shot	62.77%	39.29%	86.24%	22.73%	15.99%
llama3.3	Chain of Thought	54.81%	27.36%	82.27%	12.83%	8.38%
llama3.3	Role Prompt	54.86%	15.09%	94.63%	14.68%	14.29%
llama3.3	Step Back Prompt	57.70%	33.02%	82.38%	15.35%	10.00%
llama3.3	Thread of Thought	95.20%	96.23%	94.18%	65.38%	49.51%
llama3.3	Zero Shot	54.33%	21.70%	86.97%	12.71%	8.98%
o1-mini	Chain of Thought	61.98%	29.32%	94.63%	27.56%	26.00%
o1-mini	Role Prompt	63.54%	29.55%	97.53%	35.14%	43.33%
o1-mini	Step Back Prompt	62.08%	29.55%	94.62%	27.66%	26.00%
o1-mini	Thread of Thought	92.18%	87.40%	96.95%	74.00%	64.16%
o1-mini	Zero Shot	60.42%	25.38%	95.46%	25.78%	26.19%

Analysis and Conclusions

Self-Critique Models Statistical Test: The performance comparison between GPT-3.5 & GPT-3.5-turbo and GPT-4 & GPT-4o mod els yielded the following confidence intervals: for GPT-3.5 & GPT-3.5-turbo, the confidence interval for True Positives (TP) ranged from 0.0125 to 0.0248, while for False Positives (FP), it ranged from 0.0624 to 0.0862. For GPT-4 & GPT-4o, the confidence interval for TP was between 0.0219 and 0.0375, and the FP confidence interval ranged from 0.0870 to 0.1147. A chi-squared test was conducted to assess the differences in performance between the two groups. The resulting chi-squared statistic was 0.2789, with a p-value of 0.597 and 1 degree of freedom. The null hypothesis is that there is no significant difference between group 1 (GPT-3.5 & GPT-3.5-turbo) and group 2 (GPT-4 & GPT-4o). Since the p-value exceeds the conventional threshold of 0.05, we fail to reject the null hypothesis and conclude that there is no statistically significant difference in performance between the two models.

Impact of Prompting Techniques Statistical Test: The comparison between the Thread of Thought (ThoT) technique and all other prompting techniques yielded the following confidence intervals: for the ThoT technique, the confidence interval for True Positives ranged from 0.0519 to 0.0618, while for False Positives, it ranged from 0.0673 to 0.0783. For all other techniques, the confidence interval for TP was between 0.0144 and 0.017, and the FP confidence interval ranged from 0.0776 to 0.0833. A chi-squared test was performed to examine the difference in performance between these two groups. The resulting chi-squared statistic was 354.51, with a p-value of 4.42e-79 and 1 degree of freedom. The null hypothesis is that there is no significant difference in performance between the ThoT technique and all other techniques. Given the extremely low p-value, which is below 0.05, we reject the null hypothesis and conclude that there is a statistically significant difference in performance between the Thread of Thought technique and all other techniques.

Conclusion: The size of the model and the techniques used in prompting impact bias detection in LLMs. Larger models, such as GPT-4o, may inherit and potentially amplify biases due to their larger training scope. In contrast, smaller models like o1-mini may detect biases differently, owing to their reduced parameter count and narrower training scope. Overall, our findings highlight that while larger models may appear more capable, they are not always the best at identifying their own biases and will benefit from external critique and feedback, especially from smaller models, to improve accuracy. This could be because smaller models may detect bias better by relying on pattern recognition over deep reasoning. Our analysis revealed that employing the ThoT technique, which prompts the model to process information in manageable parts, significantly enhances performance, aiding in bias detection in the healthcare setting.

LLMs are continually being integrated into healthcare related tasks, as they have a large array of medically relevant skills. However, these models are highly vulnerable to producing biased responses. Our study queries models, critiques responses, and refines bias handling in clinical contexts. Furthermore, our study shows that there is a benefit to using smaller models to audit larger models for bias. This ushers in the concept of using other LLMs to audit model biases as they serve as an alternative to expensive medical professional assessment.

Future Steps and Limitations

Some limitations of this study include that we were unable to assess the bias of responses using a medical-specific model. This factor could have improved the study, as it would be valuable to have a model specifically trained to answer clinical questions. An interesting consideration is whether using LLMs to detect biases introduces bias itself. Since LLMs are trained on biased datasets, they may reflect and reinforce biases in their auditing. The effectiveness of bias detection also depends on the annotations used during the training.